Term Project: Stock Success

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Abstract

Project Stock Success incorporates nine datasets, each representing the stock of one company, with three each coming from the tech, defense, and waste industries, all combined into one dataframe. The purpose of this project is to use the past performance data of stocks across various industries to determine which industries do well during different times of recession and use this information to aid in creating our own investment strategies. We visualized the dCata through an interaction plot, which allows users to view the stocks of each industry and observe changes and correlations visually. One major takeaway is that during any sort of economic event, individual companies as a whole within each industry tend to perform roughly the same as each other, however, different industries will perform differently.

Background/Problem Definition

The dataset that we are using consists of historical stock data from nine companies across three industries: Technology, defense, and waste management. The data was sourced from platforms such as Yahoo Finance and includes daily stock prices (open, close, adjusted close), trading volume, and time-series data spanning multiple years. Since the datasets had various start dates, we created a long-term historical analysis and a common comparison window (2016-2025). The data preprocessing includes converting stock prices into percentage returns, normalizing values to a common baseline, and cleaning missing or inconsistent entries.

The overall problem explored in this project: Which industries and companies demonstrate the greatest resilience during economic downturns, and how can these patterns inform investment strategies?

Guiding Analytical Questions:

1. Which industry should one invest in during times of economic uncertainty, such as the 2008 financial crisis or the 2020 pandemic recession?
2. How do stock market returns change before, during, and after major market dips, and what patterns appear around these downturns?

3. Which sectors have achieved the highest average returns over the past 10 years, and how consistent has their performance been over time?
4. What stocks dip the most during periods of economic uncertainty? Could you bet against the market such as “short-selling”?
5. Does the type of recession affect which industries perform better?

Understanding sector resilience is critical for long-term investing strategies, risk management during volatility, and portfolio diversification decisions. In this project, we intend to provide a data-driven approach to evaluating investment behavior during uncertain economic periods. Visualizations associated with findings are displayed in the subsequent sections.

Data Visualization Strategy & Interactive Component

To explore our dataset and answer our guiding questions, we created several visualizations focusing on relationships between stock features, industry behavior, and market trends over time. These visualizations informed our feature selection for machine learning.

Figure 1. Pairwise Relationships Between Features

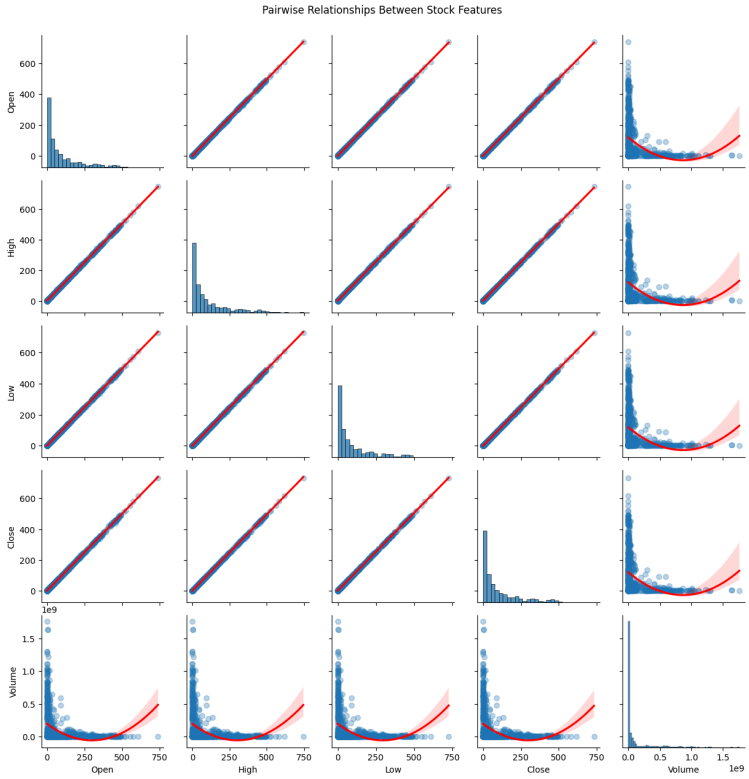


Figure 1 shows the relationships between key stock features (Open, High, Low, Close, and Volume). The price-related features are highly correlated, indicating they move closely together. In contrast, volume shows a non-linear relationship with price, suggesting trading activity is driven more by volatility than price level. This supports using derived features like returns and volatility in our machine learning model.

Figure 2. Open vs Close Prices by Industry

Figure 2 shows the relationship between opening and closing prices across industries. All industries follow a strong linear pattern, but the technology sector has a wider spread, indicating higher variability. Waste management shows the most consistent pattern, suggesting more stable price behavior. This highlights differences in volatility across industries.

Figure 3. Market Sentiment Over Time

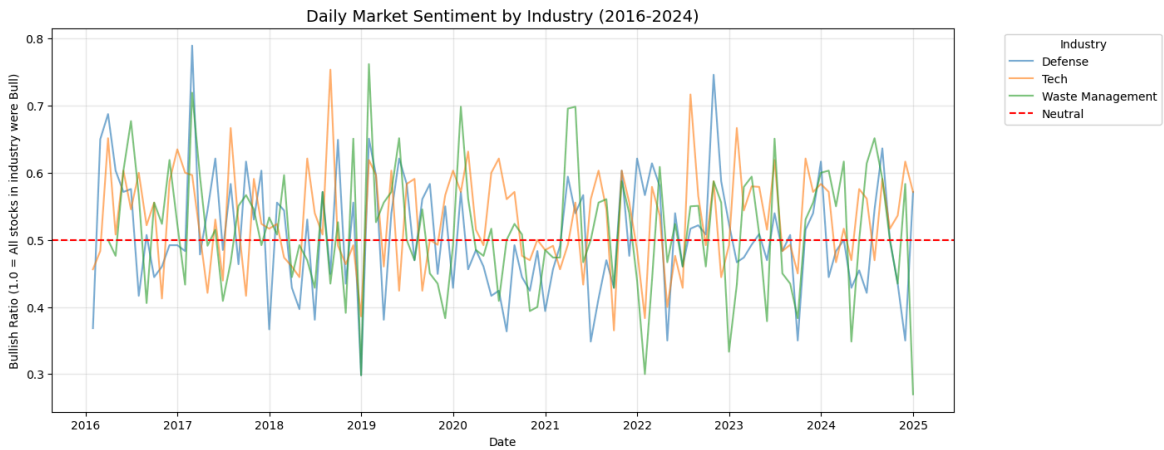


Figure 3 shows the proportion of bullish trading days over time for each industry. All industries stay close to a neutral level, meaning gains and losses are generally balanced. However, there is a clear drop during the 2020 recession, where bearish sentiment increases. Technology shows more frequent bullish periods, while defense and waste management remain more stable.

Figure 4. Stock High Prices vs Industry Average (Interactive)

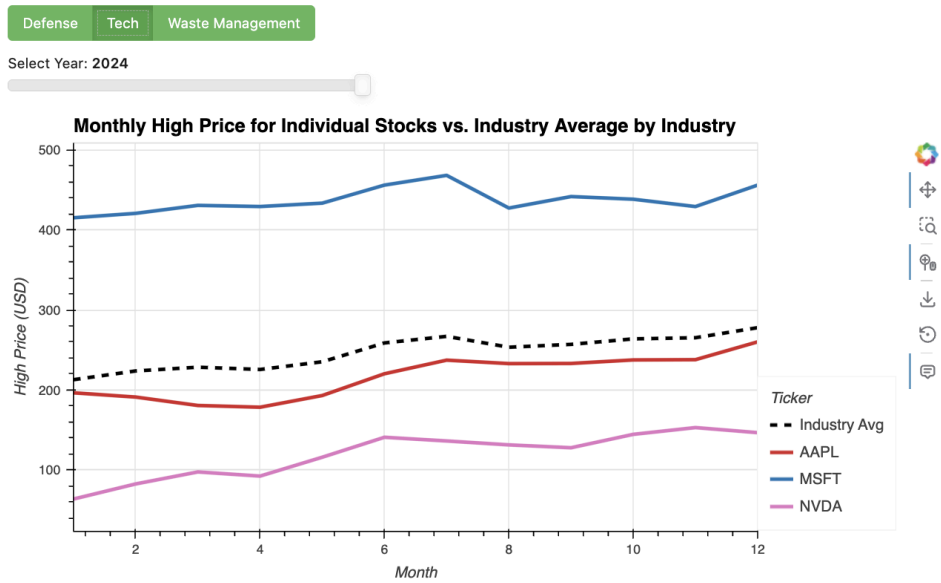


Figure 4 compares the monthly high prices of individual stocks to the industry average. It shows that certain leading stocks, particularly in the technology sector, consistently trade above the industry average, while others remain below it. Although stocks within the same industry follow similar trends, the gap between top-performing stocks and the average becomes larger during periods of strong growth.

Figure 5. Monthly Volatility by Industry (Interactive)

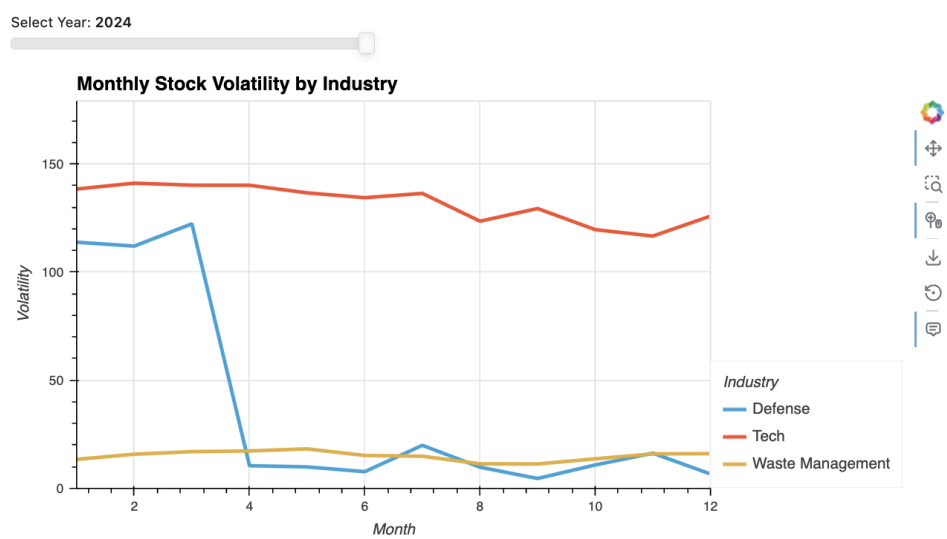


Figure 5 shows monthly stock volatility across industries. The technology sector has the highest volatility, while waste management remains the most stable. A clear spike in volatility appears across all

industries in early 2020, reflecting the market's response to the COVID-19 pandemic. This highlights differences in risk levels between industries.

Interactive Visualizations

We implemented interactive visualizations using sliders and selection tools that allow users to explore stock behavior by industry and year. Users can select specific industries and time periods to analyze individual stock performance relative to industry averages, as well as compare volatility trends. These interactions allow users to dynamically explore patterns that are not easily visible in static plots, such as identifying leading stocks or analyzing volatility during specific time periods. To run the interactive components, users can open the notebook and execute all cells. Required libraries and setup instructions are provided in the GitHub repository.

Machine Learning Pipeline & Evaluation

The primary goal of this machine learning algorithm was to leverage historical stock data and technical indicators to quantify the likelihood of a stock achieving a significant positive return. Our models were trained to predict whether a stock's price would increase by 10% or more over the subsequent 30 trading days.

Rather than providing a binary classification, our final Random Forest model provides a more granular insight. The model outputs a value between 0.0 and 1.0, which is calculated as the average of the class probabilities predicted by each individual tree in the forest. This ensemble approach allows us to see the consensus probability across all estimators.

Data Preparation & Pipeline

We prepared the dataset by splitting it into training and testing sets to evaluate model performance on unseen data. A temporal train-test split was used to preserve the time-series nature of stock data.

We engineered features such as returns, volatility, moving averages, and recession-related indicators to better capture stock behavior. Numerical features were scaled using StandardScaler, and categorical features such as industry were encoded using OneHotEncoder. Missing values were handled

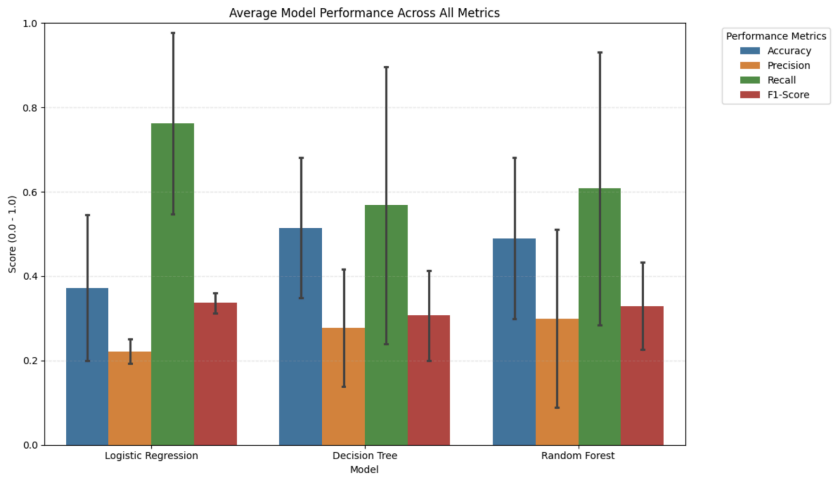
using SimpleImputer. These preprocessing steps were combined into a scikit-learn pipeline using a ColumnTransformer, ensuring consistent transformations and preventing data leakage during model training.

Model Training

We implemented three machine learning models: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was used as a baseline model, while tree-based models were used to capture nonlinear relationships in the data. To evaluate model performance, we used 5-fold cross-validation, which splits the training data into multiple subsets to ensure reliable and consistent results. Our model selection strategy was based on comparing performance metrics across models and choosing the model that provided the best overall balance.

After performing cross-validation, the model results reflected high variance across folds. This suggests that the stock data is highly sensitive to the input data folds. Comparing the models' performance after cross-validation, visualized in Figure 6, the model that outperformed the others is Random Forest.

Figure 6. Average Model Performance

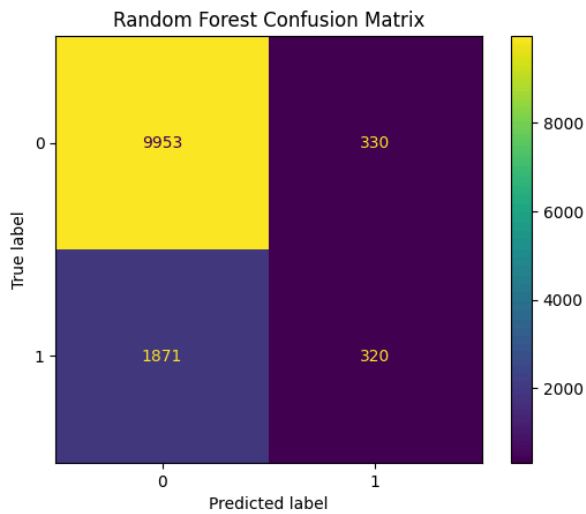


The Random Forest model achieved the highest average across all four key metrics. While Logistic Regression shows smaller variance in its precision and F1 scores, this is a result of the model consistently underperforming across all folds. As seen in figure 6, the Random Forest is the superior

choice because it achieves a higher average accuracy with similar volatility in its accuracy metrics compared to the other models.

Model Evaluation & Testing:

Figure 7. Confusion Matrix



The confusion matrix shows the performance of the Random Forest model. The model correctly classifies a large number of negative cases (0), but struggles more with positive cases (1), as seen by the higher number of misclassifications. This suggests that the model is better at identifying one class than the other, indicating some imbalance in prediction performance.

Final Interpretation

```python
def temporal_split(df, split_date='2020-01-01'):
    train = df[df['Date'] < pd.Timestamp(split_date)].copy()
    test  = df[df['Date'] >= pd.Timestamp(split_date)].copy()

    feature_cols = [
        'Open', 'High', 'Low', 'Close', 'Volume',
        'Return_1D', 'Return_5D', 'Return_20D', 'Return_60D',
        'MA_20', 'MA_50', 'MA_200',
        'Price_to_MA50', 'MA50_to_MA200',
        'Volatility_20D', 'Volatility_60D',
        'Volume_Ratio', 'RSI_14', 'BB_Position',
        'Recession_MaxDrawdown', 'Recovery_Return',
```

```
        'Recession_VolSpike', 'In_Recession',
        'Ticker', 'Industry'
    ]

    train = train.dropna(subset=feature_cols + ['Target'])
    test  = test.dropna(subset=feature_cols + ['Target'])

    X_train = train[feature_cols]
    y_train = train['Target']
    X_test = test[feature_cols]
    y_test = test['Target']

    return X_train, X_test, y_train, y_test, train, test, feature_cols
```

A key feature in our final interpretation was the splitting of our dataset into portions based on the pandemic recession of 2020. This heavily helped us determine insights leading to our conclusions.

Figure 8. Random Forest Classification Results

```
Python
ConfusionMatrixDisplay.from_predictions(y_test, y_pred)
plt.title("Random Forest Confusion Matrix")
plt.show()
```

```
              precision    recall  f1-score   support

           0       0.84      0.97      0.90     10283
           1       0.49      0.15      0.23      2191

    accuracy                           0.82     12474
   macro avg       0.67      0.56      0.56     12474
weighted avg       0.78      0.82      0.78     12474

ROC AUC: 0.6946425514400698
```

The classifier achieved an overall accuracy of 82% on the held-out test set and a weighted average F1-score of 0.78, which reflects the model's general reliability across both prediction classes. However, these numbers are heavily influenced by the dominant class, Class 0 (Do Not Buy), which represents roughly 82% of the 12,474 test samples. The model identifies these "no buy" days with high confidence, achieving a precision of 0.84 and a recall of 0.97, meaning it almost never misses a non-buy

signal. The real challenge lies in Class 1 (Buy). With only 2,191 positive cases in the test set, the model achieves a precision of 0.49 and a recall of only 0.15, resulting in an F1-score of 0.23. This means that while the model is correct about half the time when it does predict a buy signal, it catches only 1 in 7 actual buy opportunities, missing the majority.

The ROC-AUC of 0.69 confirms the model does possess some selective ability beyond random chance. It is better than a guess at ranking stocks by their likelihood of a 10% gain over 30 days, but the class imbalance inherent in the stock market, as most days do not produce 10%+ gains, makes it difficult for any model to confidently surface true buy signals without more data.

Conclusion

This project investigated which industries perform best during economic downturns to guide investment decisions. The visualizations consistently showed that technology offers the highest growth potential but also the greatest volatility, while waste management remained the most stable. The 2020 COVID-19 recession created a clear bearish dip across all three sectors, but technology recovered the fastest and most aggressively, while defense and waste management followed more gradual trajectories. These patterns directly shaped the machine learning pipeline, where recession resilience features like maximum drawdown and recovery return were engineered to capture each stock's behavior under stress. The final Random Forest model achieved 82% accuracy and a ROC-AUC of 0.69, confirming that combining technical indicators with recession-period behavior does carry meaningful predictive signals. Ultimately, the data supports a clear conclusion: waste management is the safer choice for low-risk investors during uncertain periods, while technology offers higher rewards for those with higher risk tolerance in the long run.

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